

Broad Detection Range Rhenium Diselenide Photodetector Enhanced by (3-Aminopropyl)Triethoxysilane and Triphenylphosphine Treatment

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Since the era of graphene, transition metal dichalcogenides (TMDs) with a 2D layered structure have received increasing attention as promising materials for post-silicon electronics and optoelectronics due to their exceptional electrical,^[1–3] mechanical,^[4–7] and optical^[8–10] properties. In particular, transistors fabricated on TMDs show excellent scalability of the body (or channel) thickness down to a monolayer (about 0.7 nm), consequently immunizing the short channel effect.^[11,12] Furthermore, their perfect surface without dangling bonds (thereby, no surface states or traps) leads to the suppression of scattering and enhancement of the carrier mobility, even if thickness of the TMDs is a few atoms thick.^[13] Indeed, these materials form various metal-TMD junctions, such as Ohmic or Schottky^[14,15] junctions, and also present different energy band gaps^[16] depending on the type of TMDs, thereby providing versatile electrical and optical characteristics in TMD-based devices. As a result, this approach offers many opportunities for researches in various devices, including transistors,^[17–19] photodetectors,^[20,21] light-emitting diodes,^[22–24] and solar cells.^[25–27] In addition, the carrier recombination/generation time appears to be dependent on the type of chalcogen vacancies in the TMDs, which affects the device performance (e.g., temporal response in TMD photodetectors).^[28] From this point of view, many recent studies have focused on various TMDs with the common formula MX₂, where M represents a transition metal (Mo, W, V, Nb, Ta, Ti, Zr, Hf, and Re) and X indicates a chalcogen (Se, S, and Te).^[29] However, in comparison to the intensively studied conventional TMDs, such as MoS₂, WS₂, and WSe₂,^[30–34] there are few reports on both transistors and photodetectors fabricated on other TMDs,^[35,36] consequently limiting the performance improvement and applicability of TMD devices. In addition, the optimization of TMD devices through doping technique in terms of electronic and optoelectronic device performance has not yet been performed.

Here, we investigate the electronic and optoelectronic device performance of ReSe₂-based n-channel devices and compare them to a conventional MoS₂ device, which typically operates as an n-channel transistor. In particular, we demonstrate a very high performance ReSe₂ photodetector with a broad detection range, high responsivity, and fast temporal response. This high performance is achieved by (i) suppressing the interfacial carrier scattering with (3-aminopropyl)triethoxysilane (APTES) treatment and (ii) improving the source-side contact resistance through a triphenylphosphine (PPh₃)-based n-doping technique. First, we discuss the n-doping effect of PPh₃ on both ReSe₂ and MoS₂ layers utilizing Raman spectroscopy and Kelvin probe force microscopy (KPFM) analyses. Then, the PPh₃ n-doping phenomenon is investigated in detail on ReSe₂- and MoS₂-based devices, in terms of the electronic (threshold voltage, field-effect mobility, and carrier concentration) and optoelectronic (photoresponsivity and temporal photoresponse) device performances. Additionally, we enhance the carrier mobility of the ReSe₂ device by performing an APTES treatment process on the ReSe₂/SiO₂ interface, thereby further improving its photoresponsivity.

First, to comparatively investigate the doping effects of PPh₃ on ReSe₂ and MoS₂ layers, we performed Raman spectroscopy and KPFM analyses before and after PPh₃ doping. The TMD layers were prepared on a SiO₂/Si substrate by a mechanical exfoliation method. For the Raman analysis, we prepared ten different samples and measured at three different points in each sample. As shown in Figure 1a, the pristine ReSe₂ sample (gray solid line) had two peaks near 125 and 160 cm⁻¹, which indicate E_g-like and A_g-like Raman peaks, respectively.^[37] Because of the low symmetry from the distorted triclinic structure,^[38] there were many Raman peaks between 100 and 300 cm⁻¹ (Figure S1, Supporting Information), but precise E_{2g} and A_{1g} peaks were not observed unlike other conventional TMDs such as MoS₂ and WSe₂.^[39] After applying PPh₃ doping, we observed a slight red-shift in the E_g-like and A_g-like Raman peaks (dotted red line) and the peak shift values are plotted in Figure 1b (including error bars). This red-shifting phenomenon of the Raman peaks (E_g-like: -0.78 cm⁻¹ and A_g-like: -1.29 cm⁻¹) seems to be attributed to the n-doping of ReSe₂. For reference, previous Raman studies on n-doped TMDs also reported red-shifting phenomenon in Raman peaks.^[40,41] This n-doping effect by the PPh₃ was also confirmed in the MoS₂ samples, as shown in Figure 1c,d. The obtained red-shifted values were respectively -0.78 and -0.87 cm⁻¹ for the E_{12g} and A_{1g} Raman peaks after performing the PPh₃ doping process on

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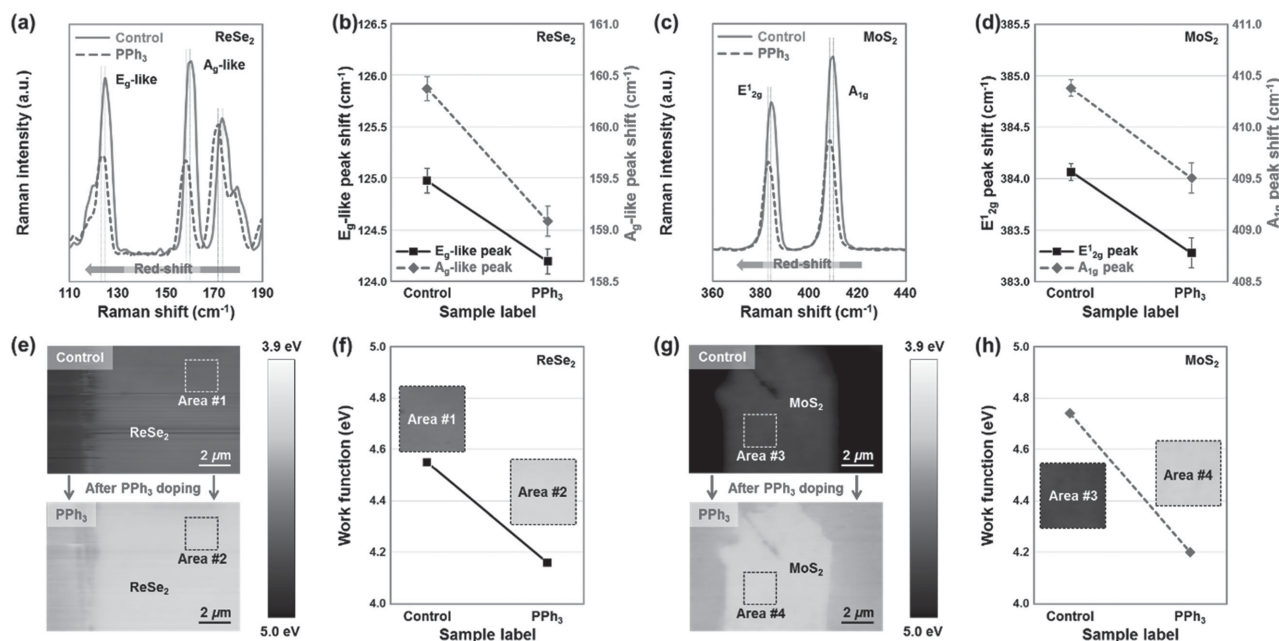


Figure 1. Characterization of ReSe₂ and MoS₂ layers before/after PPh₃ doping. Raman spectra of a) ReSe₂ and c) MoS₂ layers before (solid gray lines) and after PPh₃ doping (dotted red lines). Extracted Raman peak shift values of b) ReSe₂ and d) MoS₂ layers doped by PPh₃. KPFM mapping images and work function values were respectively obtained on the e,f) ReSe₂ and g,h) MoS₂ surfaces before and after PPh₃ doping.

MoS₂. Meanwhile, the full width at half maximum values of the A_g-like (ReSe₂) and A_{1g} (MoS₂) peaks showed a broadening trend after PPh₃ n-doping in both the ReSe₂ (from 3.93 to 4.27 cm⁻¹) and MoS₂ (from 5.35 to 5.61 cm⁻¹) layers, while maintaining relatively small changes in the E_g-like (ReSe₂) and E_{12g} (MoS₂) peaks. These results are in agreement with a previously reported study, where the electrostatic n-doping of MoS₂ presented the stronger electron-phonon coupling in the A_{1g} mode than the E_{2g} mode.^[42] In addition, the photoluminescence spectra of TMD layers (ReSe₂ and MoS₂) before and after PPh₃-doping were measured and plotted (Figure S2, Supporting Information), which also indicate the n-doping phenomenon on the TMD layers. In order to support the n-type doping phenomenon of PPh₃ on the ReSe₂ and MoS₂ layers, we additionally performed KPFM analysis. Before the measurement, KPFM tip was calibrated on a highly oriented pyrolytic graphite (HOPG) surface. We then obtain the work function (W) values from the contact potential difference (ΔV_{CPD}) between the KPFM tip and TMD surface, which is specifically explained in experimental section. Figure 1e,g respectively shows the work function mapping images taken on the ReSe₂ and MoS₂ surfaces before and after PPh₃ doping. After applying PPh₃, the brighter images were observed in both ReSe₂ and MoS₂ surfaces, indicating that the work functions of the TMDs were decreased due to the n-doping effect of PPh₃. The average work function values were extracted from specific area (inside of dotted line) of the ReSe₂ ($W_{\text{ReSe2Control}}$: 4.55 eV and $W_{\text{ReSe2PPh3}}$: 4.16 eV) and MoS₂ ($W_{\text{MoS2Control}}$: 4.74 eV and W_{MoS2PPh3} : 4.20 eV) surfaces, as shown in Figure 1f,h, where the decreasing trend of the work function values after the PPh₃ doping supports the n-type doping phenomenon of TMDs predicted by the Raman analysis (Figure 1a-d). In addition, small root-mean-square (RMS) values (15–19 meV for pristine and 12–16 meV for PPh₃-doped

TMD layers) were observed, meaning that the TMDs were uniformly n-doped by PPh₃.

Next, in order to verify the n-doping effects of PPh₃ in terms of the ReSe₂ and MoS₂ device performance, we fabricated back-gated field-effect transistor (FET) devices using the TMD layers with similar thicknesses (Figure S3, Supporting Information) and then performed electrical measurements (I_D - V_G and I_D - V_D). Figure 2a presents the schematic diagram and optical image of the PPh₃-doped TMD device (both the channel length and width were 5 μm) and the illustration of the n-doping mechanism by PPh₃ at the PPh₃/TMD interface. When the PPh₃ layer is coated on the TMD devices, the lone electron pair of the phosphorus atom in the PPh₃ molecule transfers negative charges to the TMD layers, subsequently resulting in an n-doping phenomenon. Figure 2b,c shows the I_D - V_G curves of the ReSe₂ and MoS₂ devices before and after PPh₃ doping, where the applied drain voltage (V_{DS}) was 5 V. After applying the PPh₃ n-doping technique, we observed a 9.79-fold increase in the on-current (I_{on}) (from 2.36×10^{-7} to 2.31×10^{-6} A μm^{-1}) and a 7.08-fold increase (from 2.81×10^{-6} to 1.99×10^{-5} A μm^{-1}) in the ReSe₂- and MoS₂-based FETs, respectively, where the on-current was obtained at $V_{GS} - V_{TH} = 50$ V. The threshold voltage (V_{TH}) was negatively shifted ($\Delta V_{TH\text{ReSe2}}$: -10.64 V and $\Delta V_{TH\text{MoS2}}$: -8.58 V) after the n-doping and the extracted values are plotted in Figure 2d. Because of the n-type doping phenomenon, which transfers electrons from PPh₃ to the TMD layers, the Fermi level of TMD is up-shifted and the TMD energy band is down-shifted, thereby decreasing the effective electron barrier height (Φ_{eff}) at the source-TMD junctions, as described in Figure 2e. Consequently, the electron injection probability from the source to TMD (channel) is expected to increase in both the ReSe₂ and MoS₂ devices. In addition, the I_D - V_D characteristics were also measured and plotted with increasing V_{GS} (Figure S4,

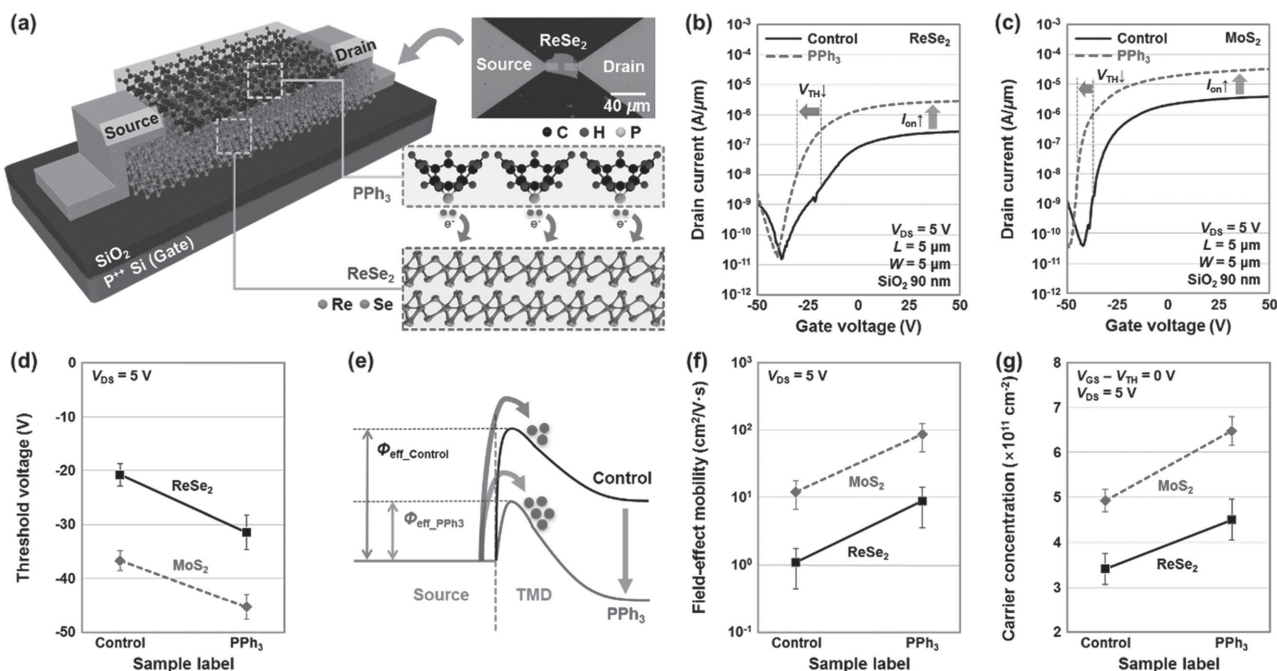


Figure 2. Characterization of PPh₃-doped ReSe₂ and MoS₂ transistors. a) Schematic diagram and optical image of back-gated (90 nm thick SiO₂ gate dielectric) ReSe₂ and MoS₂ transistor devices doped by PPh₃ and a descriptive illustration for the n-doping mechanism by PPh₃ at the PPh₃/TMD interface. I_D - V_G characteristics of b) ReSe₂ and c) MoS₂ transistor devices before and after PPh₃ doping. d) Extracted threshold voltage shifts ($\Delta V_{TH} = V_{THPPh3} - V_{THControl}$) of the TMD devices after PPh₃ doping. e) Energy band diagram of the source-TMD junction before and after PPh₃ doping. Calculated f) field-effect mobility and g) carrier concentration at $V_{GS} = V_{TH}$ in TMD transistors before and after PPh₃ doping.

Supporting Information), which well supports the n-doping phenomenon on the TMD devices. Here, we observed a couple of differences in the electronic performance of the ReSe₂- and MoS₂-based FETs where the ReSe₂ FET needed a higher gate voltage (V_{GS}) bias to turn-on the device ($V_{THReSe2} > V_{THMoS2}$) and had a smaller on-current level ($I_{onReSe2} < I_{onMoS2}$) than the MoS₂ FET in both the pristine and PPh₃-doped devices. These results are attributable to the difference of the electron barrier height at the source metal-TMD junctions. The Fermi level pinning phenomenon at the Ti-TMD (ReSe₂ or MoS₂) junctions seems to cause a higher electron barrier height in the Ti-ReSe₂ junction. As shown in Figure 2f,g, the field-effect mobility (μ_{FE}) and carrier concentration (n) were extracted, and we compared the performance of the ReSe₂ and MoS₂ devices before and after the PPh₃ n-doping considering these parameters. As a result of successful n-doping, the μ_{FE} values of both the ReSe₂- and MoS₂-based devices improved from 1.09 to 8.85 cm² V⁻¹ s⁻¹ and from 12.08 to 86.04 cm² V⁻¹ s⁻¹, respectively. The I_D - V_G curves on a linear scale clearly show enhanced μ_{FE} values after n-doping with PPh₃ (Figure S5, Supporting Information). This enhancement of μ_{FE} was attributed to the increased electron injection probability from the source to the TMD channel. Although the μ_{FE} of the TMD FETs improved after applying PPh₃, μ_{FE} in ReSe₂ was still smaller than that in MoS₂ due to the higher contact resistance at the Ti-ReSe₂ junction. Furthermore, in Figure 2g, we also observed a similar increase of the carrier concentration (Δn_{ReSe2} : 1.09×10^{11} cm⁻² and Δn_{MoS2} : 1.56×10^{11} cm⁻²) in both TMD devices after PPh₃ n-doping, where the carrier concentrations were extracted at $V_{GS} - V_{TH} = 0$ V. In particular, the ReSe₂ device showed smaller carrier concentrations than MoS₂ before

and after the PPh₃ n-doping, supporting the notion that its Fermi level was pinned further down from the conduction band edge at the source-TMD junction. In terms of the doping concentration, this n-doping phenomenon was similar to previous doping techniques of TMD in the non-degenerate regime, such as n-doping by Cs₂CO₃^[43] (approximately 0.5 – 3.5×10^{11} cm⁻²), n-doping by P₂O₅^[44] (approximately 0.4 – 4.5×10^{11} cm⁻²), and p-doping by OTS^[45] (approximately 2.1 – 5.2×10^{11} cm⁻²). We also compared our work with the previously reported organic material-based doping techniques for TMD devices, in terms of the doping type and the achieved concentration (Table S1, Supporting Information). Additionally, the stability of the PPh₃-based n-doping was investigated in terms of variations of the electrical parameter (V_{TH}) (Figure S6, Supporting Information). Even though the PPh₃-doped devices were exposed to air for 14 d, we confirmed very small shifts of V_{TH} ($\Delta V_{THReSe2}$: 6.65% and ΔV_{THMoS2} : 7.58% after 14 d), which indicate relatively good stability compared to previous doping techniques performed on TMD devices.^[45,46]

In order to investigate the PPh₃ n-doping effects in terms of optoelectronic device performance, we fabricated ReSe₂ and MoS₂ photodetectors and performed electrical measurements (I_D - V_G) under dark and laser illuminated conditions. Figure 3a presents the schematic diagram of the PPh₃-doped TMD photodetector and the energy band diagrams for the source-TMD junctions (Ti-ReSe₂ and Ti-MoS₂) under laser illuminated conditions, where ReSe₂ (1.0 eV^[35]) has a smaller optical band gap than MoS₂ (1.3 eV^[47]) and Ti-ReSe₂ junction shows a higher effective electron barrier height than Ti-MoS₂. Because the effective electron barrier height at the metal-TMD

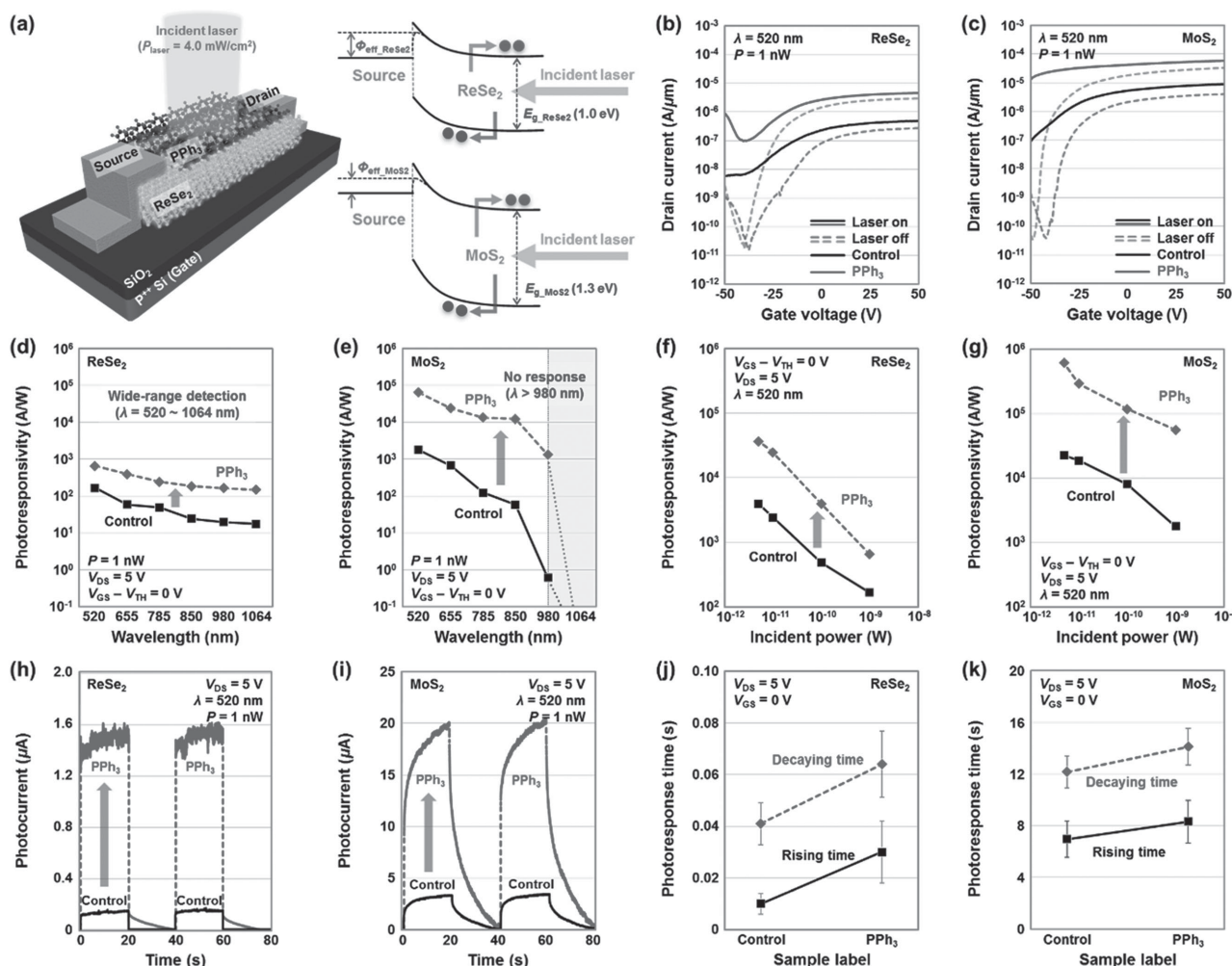


Figure 3. Characterization of PPh₃-doped ReSe₂ and MoS₂ photodetectors. a) Schematic diagram of PPh₃-doped TMD photodetector under laser illumination ($P_{\text{laser}} = 4 \text{ mW cm}^{-2}$) and energy band diagrams of source-TMD junctions (Ti-ReSe₂ and Ti-MoS₂). I_D - V_G characteristics of b) ReSe₂ and c) MoS₂ photodetectors before (dark line) and after PPh₃ doping (red line) under both dark (dotted line) and illuminated (solid line) conditions. Photoresponsivity as a function of the wavelength extracted at $V_{GS} = V_{TH}$ in the d) ReSe₂ and e) MoS₂ photodetectors. Photoresponsivity as a function of the incident laser power obtained at $V_{GS} = V_{TH}$ before and after PPh₃ doping in the f) ReSe₂ and g) MoS₂ photodetectors. Temporal photoresponse curves and extracted rising/decaying times h,i) ReSe₂ and i,k) MoS₂ photodetectors obtained at $V_{GS} = 0 \text{ V}$.

junctions decreased after PPh₃ n-doping, the contact resistance is expected to be reduced, eventually increasing the electric field in the junction interface and the collection probability for carriers inside the TMDs, as shown in Figure 3b,c. After PPh₃ n-doping on the ReSe₂ photodetector, the photocurrent ($I_{ph} = I_{\text{laseron}} - I_{\text{laseroff}}$) increased from 5.05×10^{-8} to $1.96 \times 10^{-7} \text{ A } \mu\text{m}^{-1}$, which was extracted at the bias condition, $V_{GS} - V_{TH} = 0 \text{ V}$ and $V_{DS} = 5 \text{ V}$. In the case of the MoS₂ photodetector, the photocurrent increased from 5.42×10^{-7} to $1.94 \times 10^{-5} \text{ A } \mu\text{m}^{-1}$. In addition, the difference between the photocurrent levels of the ReSe₂ and MoS₂ photodetectors ($I_{ph\text{ReSe}_2} < I_{ph\text{MoS}_2}$) also supports the previous analysis in which the Ti-ReSe₂ junction has a higher electron barrier and contact resistance than MoS₂ (Figure 2). We then extracted the photoresponsivity before and after the PPh₃ n-doping at $V_{GS} - V_{TH} = 0 \text{ V}$ and compared the values depending on the wavelength (Figure 3d,e) and incident power (Figure 3f,g) of the laser. Under laser illumination, the ReSe₂ photodetector showed similar photoresponsivity values

for all of the wavelengths, indicating that the cutoff wavelength (λ_{cutoff}) is more than 1064 nm. On the other hand, in the case of the MoS₂ photodetector, a significant degradation of the photoresponsivity was observed above a wavelength of 980 nm (no response at 1064 nm). The broadband photodetection of the ReSe₂-based photodetector from the visible to near-infrared region resulted from its relatively small optical band gap compared to MoS₂, as depicted in Figure 3a. In particular, the cutoff wavelength value observed in this ReSe₂ photodetector ($\lambda_{\text{cutoffReSe}_2} > 1064 \text{ nm}$) is much higher than the λ_{cutoff} of previously studied TMD photodetectors based on MoS₂^[48] and WSe₂,^[49] demonstrating the promising use of ReSe₂ in optoelectronic applications for broadband photodetection. Additionally, after applying the PPh₃ n-doping technique, the photoresponsivity values improved over the entire wavelength range due to the increased electric field in the TMD region as a result of the reduced contact resistance at the metal-TMD junction. In sequence, in order to analyze the doping effects on the TMD

photodetectors according to the incident power of lasers, the photoresponsivity values as a function of the incident power (5, 10, 100, and 1000 pW) were extracted and are plotted in Figure 3f,g. In both the ReSe₂ and MoS₂ photodetectors, we observed exponential improvement of the photoresponsivity as the incident power of the laser decreased due to the suppressed scattering phenomenon among the photogenerated carriers.^[50] Then, the maximum photoresponsivity values were obtained under 5 pW laser illumination in the doped ReSe₂ and MoS₂ photodetectors, where the extracted values were 3.68×10^4 and 6.13×10^5 A W⁻¹, respectively. The lower photoresponsivity values of the ReSe₂ device (compared to the MoS₂ device) are attributed to the higher contact resistance at the metal-TMD junction, which weakens the electric field distributed in the TMD region. Furthermore, the temporal photoresponse characteristics were investigated in the undoped (control) and PPh₃-doped TMD (ReSe₂ and MoS₂) photodetectors with laser switching cycles of 20 s. The extraction method for the rising time (τ_r) and decaying time (τ_d) is specifically explained (Figure S7, Supporting Information). Figure 3h,i presents the temporal photoresponse curves of the ReSe₂- and MoS₂ photodetectors, respectively, where the enhancement of the photocurrents by PPh₃ n-doping was confirmed again. In particular, in the case of the ReSe₂ photodetector (Figure 3h), a sharp increase or decrease was observed at the rising and decaying edges of the temporal photoresponse curves, indicating the much faster photoswitching speed of the ReSe₂-based device compared to the MoS₂ device showing relatively slow changes. The extracted photoresponse times are plotted in Figure 3j,k. As expected, the ReSe₂ photodetector showed much shorter rising and decaying times compared to the MoS₂ photodetector (τ_r/τ_d : 0.01/0.041 s for ReSe₂ and 6.97/12.2 s for MoS₂). The difference in photoswitching speed of the ReSe₂- and MoS₂-based devices seems to be caused by the kind of chalcogen vacancies in the TMD materials, where selenium vacancies serving as traps possess much shorter recombination and generation times for carriers than sulfur vacancies. This result is consistent with previous studies of sulfide.^[51,52] and selenide-based^[28,49] TMD photodetectors, where a much shorter photoresponse time was obtained in the selenide-based TMD devices (e.g., about 40 μ s for both rising and decaying edges in WSe₂).^[53] Exceptionally, when a photodetector was fabricated on tin-based chalcogenides (SnS₂ or SnSe₂), its rising and decaying times were very short, as much as 5–7 μ s.^[54] In addition, the photoresponse (both rising and decaying) times slightly increased after PPh₃ n-doping in both the ReSe₂ (τ_r : 0.01 $\rightarrow$ 0.03 s and τ_d : 0.041 $\rightarrow$ 0.064 s) and MoS₂ (τ_r : 6.97 $\rightarrow$ 8.31 s and τ_d : 12.2 $\rightarrow$ 14.1 s) photodetectors. This phenomenon is probably attributable to the variation of the depletion width at the metal-TMD junctions. After applying PPh₃ n-doping, the depletion width at the metal-TMD junctions is expected to be shortened, which subsequently makes it difficult to collect the photogenerated carriers in the TMD region.

Overall, the PPh₃-doped MoS₂ photodetector showed a high photoresponsivity value ($R_{\text{MoS}_2\text{PPh}_3}$: 6.13×10^5 A W⁻¹), but also very long photoresponse times (τ_r : 8.31 s and τ_d : 14.1 s) and a limited detection range. Whereas in the case of ReSe₂ photodetector, the photoresponsivity ($R_{\text{ReSe}_2\text{PPh}_3}$: 3.68×10^4 A W⁻¹) was enhanced after applying the PPh₃ n-doping technique, while maintaining a broad photodetection range (from visible

to near-infrared) and short photoresponse times (τ_r : 30 ms and τ_d : 64 ms). However, even after the PPh₃ doping, the photoresponsivity value of the ReSe₂ device was still lower than that of the MoS₂ device due to its higher contact resistance at the Ti-ReSe₂ junction, as mentioned above in Figures 2 and 3. Therefore, in order to further improve the photoresponsivity of the ReSe₂ photodetector and eventually achieve a comparable value to that of the MoS₂ photodetector, APTES, which is a self-assembled monolayer material, was inserted between the ReSe₂ and SiO₂ layers. Because the APTES layer induces the n-doping effect^[55] and blocks the charge puddle effect^[56,57] on the SiO₂ layer under the channel region, it can further reduce the contact resistance at the metal-TMD junction and also improve the carrier mobility of TMD FETs. Figure 4a shows the schematic diagram of the PPh₃-doped ReSe₂ device with an APTES layer on the SiO₂ substrate as well as the detailed description of the transfer process onto an APTES-treated substrate that was prepared in advance. Here, we used a wet transfer method to fabricate the APTES-treated ReSe₂ device (Figure S9, Supporting Information) and investigated the difference of the device performance on the same TMD layer before and after inserting the APTES layer. After applying the APTES layer, negative charges from the amine functional groups ($-\text{NH}_2$) in the APTES layer seem to induce the accumulation of hole carriers at the ReSe₂/APTES interface, thereby showing n-type doping phenomenon on ReSe₂.^[55] Furthermore, the suppressed charge puddle effect on the SiO₂ layer is also expected to enhance the carrier mobility of the ReSe₂ device by separating the ReSe₂ channel region from the charged surface of SiO₂.^[57] As shown in Figure 3b,c, the electronic performance of the PPh₃-doped ReSe₂ device was measured before and after inserting an APTES layer, where the device performance improved in terms of the on-current (from 2.31×10^{-6} to 8.45×10^{-6} A μm^{-1} at $V_{\text{GS}} - V_{\text{TH}} = 50$ V), field-effect mobility (from 8.85 to 33.45 cm² V⁻¹s⁻¹), and carrier concentration (from 4.50×10^{11} to 5.25×10^{11} cm⁻²). The $I_{\text{D}}-V_{\text{G}}$ curves were also plotted on a linear scale (Figure S10, Supporting Information) to clearly present the improvement in electronic performance of the APTES-inserted ReSe₂ device. These changes are potentially attributed to the further decreased effective electron barrier height at the Ti-ReSe₂ junction by applying the APTES layer, in addition to the barrier height reduction after PPh₃-based n-doping (Figure 2). Next, to investigate the optoelectronic device performance, we extracted the photoresponsivity values of the ReSe₂ photodetector with and without the APTES layer and compared the values as a function of the wavelength (Figure 4d) and incident power (Figure 4e) of the laser. In the case of the 520 nm laser, the photoresponsivity was enhanced by a factor of 27.0 (from 6.53×10^2 to 1.76×10^4 A W⁻¹) after combining the APTES treatment and PPh₃ doping technique. A similar improvement was also confirmed in the other cases with different laser wavelengths, as shown in Figure 4d. Figure 4e shows the photoresponsivity values before and after the APTES treatment with decreasing incident power of the laser, where the extracted maximum value was approximately 1.18×10^6 A W (with the APTES layer) under 5 pW laser exposure due to the reduced scattering probability of photogenerated carriers. This high photoresponsivity value is slightly higher than that of the PPh₃-doped MoS₂ photodetector ($R_{\text{MoS}_2\text{PPh}_3}$: 6.13×10^5 A W⁻¹). Such an enhanced

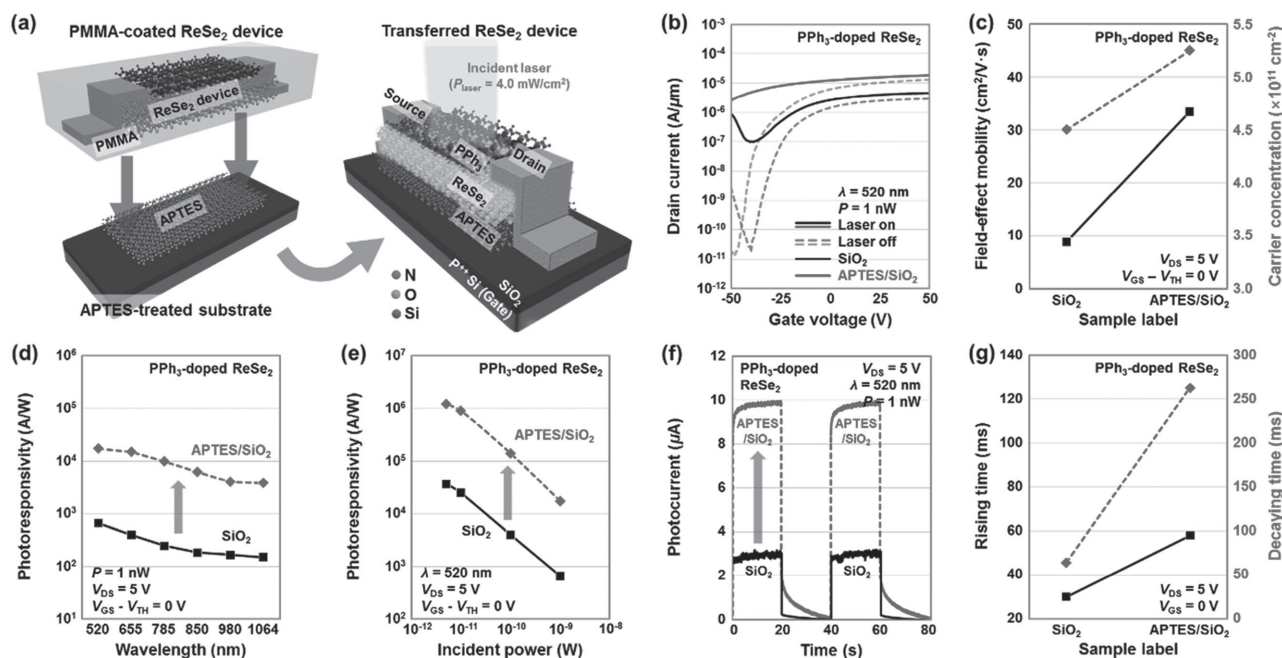


Figure 4. Characterization of PPH₃-doped ReSe₂/APTES photodetectors. a) Schematic diagrams of (left) transferring process onto an APTES-treated substrate and (right) PPH₃-doped ReSe₂ photodetector after the transferring process. b) I_D - V_G curves of PPH₃-doped ReSe₂ photodetectors before (dark line) and after inserting the APTES layer (red line) under both dark (dotted line) and illuminated (solid line) conditions. c) Calculated field-effect mobility and carrier concentration extracted at $V_{GS} = V_{TH}$ in PPH₃-doped ReSe₂ photodetectors without and with an APTES layer. Photoresponsivity as functions of the d) wavelength and e) incident power of the laser, which were extracted at $V_{GS} = V_{TH}$ in the PPH₃-doped ReSe₂ before and after transferring onto the APTES/SiO₂ substrate. f) Temporal photoresponse curves and g) extracted rising/decaying times of the PPH₃-doped ReSe₂ photodetector obtained at $V_{GS} = 0$ V.

photoresponsivity value was achieved by (i) the reduced contact resistance, which increased the electric field in the ReSe₂ channel region, and (ii) the additionally increased carrier mobility by the separation of the ReSe₂ channel and charged substrate, subsequently enhancing the collection probability of the photogenerated carriers in the ReSe₂ channel region. The temporal photoresponse characteristics shown in Figure 4f,g ($V_{DS} = 5$ V and $V_{GS} = 0$ V) clearly show that the photocurrent of the PPH₃-doped ReSe₂ photodetector improved after inserting the APTES layer while maintaining a faster photoresponse speed compared to the MoS₂-based device (above Figure 3k). In addition, slightly increased photoresponse times (τ_r : 30 → 58 ms and τ_d : 64 → 263 ms) were observed after applying the APTES layer due to the narrowed depletion width at the Ti-ReSe₂ junction by APTES-based additional n-type doping, but the values are still lower by about two orders of magnitude than the MoS₂ device (Figure 3j,k). In addition, all the extracted parameters in this work were summarized in Table S2 in the Supporting Information, where a comparison in terms of the electronic and optoelectronic performance is possible.

In conclusion, we demonstrated a ReSe₂-based photodetector with a high responsivity, fast switching speed, and broad detection range by (i) PPH₃ n-doping of the ReSe₂ channel and (ii) APTES layer insertion into the ReSe₂/SiO₂ interface. In particular, the capability of wide-range detection from visible to near-infrared wavelengths opens the door for application of ReSe₂ for broadband photodetection, which is limited in other conventional TMDs due to their variable optical band gap. The n-doping effect was based on the negative charge transfer

phenomenon from the electron lone pair of phosphorus atoms in PPH₃ molecules, and an n-doping concentration (Δn) of the TMD layers between 1.09×10^{11} and 1.56×10^{11} cm⁻² (non-degenerate regime) was obtained. We also investigated the PPH₃ doping effects in terms of the electronic (on-current, threshold voltage shift, field-effect mobility, and carrier concentration) and optoelectronic device performances (photoresponsivity and photoresponse times) by comparing ReSe₂ and MoS₂ devices. By performing the PPH₃-based n-doping process, we achieved an ReSe₂ photodetector with an enhanced photoresponsivity ($3.97 \times 10^3 \rightarrow 3.68 \times 10^4$ A W⁻¹) while maintaining a broad photodetection range (possible above 1064 nm) and fast photo-switching speed (τ_r : 30 ms and τ_d : 64 ms), but the photoresponsivity value of the ReSe₂ device was still lower than that of MoS₂ device (6.13×10^5 A W⁻¹). The slightly lower photoresponsivity was then further improved by applying an APTES layer into the ReSe₂/SiO₂ interface because it can suppress the scattering phenomenon at the channel region and induce additional n-doping via its functional group ($-NH_2$). Consequently, the maximum photoresponsivity of 1.18×10^6 A W⁻¹ was achieved in the ReSe₂ photodetector, which was higher than that of the MoS₂ photodetector, where a broad detection range and short photoresponse times were also maintained.

Experimental Section

PPH₃ n-Doping of TMD: For preparation of the PPH₃ solution with a PPH₃ concentration of 7.5 wt%, 0.75 g of PPH₃ was dissolved in 9.25 g

of toluene. For doping of TMD layers, the PPh_3 solution was spin-coated on the prepared TMD samples at 1000 rpm for 30 s (5 s acceleration). The samples were then annealed at 150 °C for 10 min to remove the remaining solvent. The ReSe_2 crystals were commercially purchased and thin ReSe_2 layer was mechanically exfoliated.

Raman and KPFM Analyses of PPh_3 -Doped TMD: The characteristics of ReSe_2 and MoS_2 (the flakes were mechanically exfoliated and transferred onto the SiO_2/Si substrate using adhesive tape) were investigated by Raman spectroscopy (Alpha300 M+, WITec) before and after PPh_3 n-doping. Raman spectroscopy with an excitation wavelength of 532 nm was used with a laser beam size of 0.7–0.9 μm and an instrument spectral resolution less than 0.9 cm^{-1} . The integration time was 5 s and a spectrometer with 1800 grooves mm^{-1} was employed for the test. For the KPFM measurement, a platinum/iridium (Pt/Ir) coated Si tip was used and the tip was calibrated on a HOPG surface. First, this study calculated the work function of the KPFM tip ($W_{\text{tip}} - W_{\text{HOPG}} = \Delta V_{\text{CPDHOPG}}$) using the well-known work function of the HOPG ($W_{\text{HOPG}} = 4.6$ eV) and the contact potential difference between tip and HOPG ($\Delta V_{\text{CPDHOPG}} = 324$ meV). The work function of the TMD layers with the calculated W_{tip} (4.92 eV) value and the measured ΔV_{CPDTMD} between KPFM tip and TMD surface ($W_{\text{tip}} - W_{\text{TMD}} = \Delta V_{\text{CPDTMD}}$) was then found.

Fabrication of PPh_3 -Doped TMD Electronic and Optoelectronic Devices: The ReSe_2 and MoS_2 layers were mechanically exfoliated and transferred to the 90 nm thick SiO_2 on a heavily boron-doped Si substrate via adhesive tape (224SPV, Nitto). The remaining adhesive tape residue was removed by immersing the samples in an acetone bath for 3 h. For the fabrication of back-gated TMD devices (transistors and photodetectors), source/drain electrode regions (channel length and width were 5 μm) were patterned on the TMD/ SiO_2/Si samples by an optical lithography process, followed by the deposition of 10 nm thick titanium (Ti) and 40 nm thick palladium via an e-beam evaporator. Additionally, the fabricated TMD devices were transferred to the APTES/ SiO_2/Si substrate by a wet transfer method using poly(methyl methacrylate) supporting layers (Figure S9, Supporting Information).

Characterization of the PPh_3 -Doped TMD Electronic Device: The fabricated devices were electrically analyzed through current–voltage measurements ($I_{\text{D}}-V_{\text{G}}$ and $I_{\text{D}}-V_{\text{D}}$) by a Keysight B2912A precision source/measure unit. The threshold voltage (V_{TH}), carrier concentration (n), and field-effect mobility (μ_{FE}) were calculated from the measured data. Here, all drain currents (I_{D}) were normalized by the channel width (W). The carrier concentration and field-effect mobility were respectively extracted using $n = I_{\text{D}}L/qW\mu_{\text{FE}}V_{\text{D}}$ and $\mu_{\text{FE}} = L/(WV_{\text{D}}C_{\text{OX}}) \times (\partial I_{\text{D}}/\partial V_{\text{G}})$, where L and W are the length and width of the channel, respectively, q is the electron charge, and the gate oxide capacitance per unit area C_{OX} is $\epsilon_{\text{OX}} \times \epsilon_0/t_{\text{OX}}$.

Characterization of the PPh_3 -Doped TMD Optoelectronic Device: The PPh_3 -doped ReSe_2 and MoS_2 photodetector devices were investigated by performing electrical measurements ($I_{\text{D}}-V_{\text{G}}$) under both dark and illuminated conditions. The light sources were dot lasers with wavelengths of 520, 655, 785, 850, 980, and 1064 nm and an optical power of 4.0 mW cm^{-2} . The photoresponsivity (R) was extracted using $R = I_{\text{ph}}/P_{\text{laser}}$ where I_{ph} is the generated photocurrent and P_{laser} is the total incident optical power, which varied at values of 5, 10, 100, and 1000 pW. The photoswitching characteristics of the PPh_3 -doped TMD photodetector were investigated under 520 nm laser illumination, which was turned on and off with a cycle of 20 s. The photoresponse times (rising and decaying times) were then extracted between 10% and 90% of the measured maximum photocurrent (I_{max}) data (Figure S7, Supporting Information).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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- [1] B. Radisavljevic, A. Radenovic, J. Brivio, V. Giacometti, A. Kis, *Nat. Nanotechnol.* **2011**, 6, 147.
- [2] X. Hong, J. Kim, S.-F. Shi, Y. Zhang, C. Jin, Y. Sun, S. Tongay, J. Wu, Y. Zhang, F. Wang, *Nat. Nanotechnol.* **2014**, 9, 682.
- [3] S. Kim, A. Konar, W.-S. Hwang, J. H. Lee, J. Lee, J. Yang, C. Jung, H. Kim, J.-B. Yoo, J.-Y. Choi, Y. W. Jin, S. Y. Lee, D. Jena, W. Choi, K. Kim, *Nat. Commun.* **2012**, 3, 1011.
- [4] P. Johari, V. B. Shenoy, *ACS Nano* **2012**, 6, 5449.
- [5] A. P. Nayak, S. Bhattacharyya, J. Zhu, J. Liu, X. Wu, T. Pandey, C. Jin, A. K. Singh, D. Akinwande, J.-F. Lin, *Nat. Commun.* **2014**, 5, 3731.
- [6] W. S. Yun, S. W. Han, S. C. Hong, I. G. Kim, J. D. Lee, *Phys. Rev. B* **2012**, 85, 033305.
- [7] M. Ghorbani-Asl, S. Borini, A. Kuc, T. Heine, *Phys. Rev. B* **2013**, 87, 235434.
- [8] O. Lopez-Sanchez, D. Lembke, M. Kayci, A. Radenovic, A. Kis, *Nat. Nanotechnol.* **2013**, 8, 497.
- [9] K. F. Mak, K. He, C. Lee, G. H. Lee, J. Hone, T. F. Heinz, J. Shan, *Nat. Mater.* **2013**, 12, 207.
- [10] A. Carvalho, R. M. Ribeiro, A. H. C. Neto, *Phys. Rev. B* **2013**, 88, 115205.
- [11] H. Liu, A. T. Neal, P. D. Ye, *ACS Nano* **2012**, 6, 8563.
- [12] Y. Yoon, K. Ganapathi, S. Salahuddin, *Nano Lett.* **2011**, 11, 3768.
- [13] V. Podzorov, M. E. Gershenson, Ch. Kloc, R. Zeis, E. Bucher, *Appl. Phys. Lett.* **2004**, 84, 3301.
- [14] H.-J. Chuang, B. Chamlagain, M. Koehler, M. M. Perera, J. Yan, D. Mandrus, D. Tomanek, Z. Zhou, *Nano Lett.*
- [15] J. Kang, W. Liu, D. Sarkar, D. Jena, K. Banerjee, *Phys. Rev. X* **2014**, 4, 031005.
- [16] A. Kumar, P. K. Ahluwalia, *Eur. Phys. J. B* **2012**, 85, 185.
- [17] T. Roy, M. Tosun, J. S. Kang, A. B. Sachid, S. B. Desai, M. Hettick, C. C. Hu, A. Javey, *ACS Nano* **2014**, 8, 6259.
- [18] W. Liu, J. Kang, D. Sarkar, Y. Khatami, D. Jena, K. Banerjee, *Nano Lett.* **2013**, 13, 1983.
- [19] J. Pu, Y. Yomogida, K.-K. Liu, L.-J. Li, Y. Iwasa, T. Takenobu, *Nano Lett.* **2012**, 12, 4013.
- [20] F. H. L. Koppens, T. Mueller, Ph. Avouris, A. C. Ferrari, M. S. Vitiello, M. Polini, *Nat. Nanotechnol.* **2014**, 9, 780.
- [21] E. Zhang, Y. Jin, X. Yuan, W. Wang, C. Zhang, L. Tang, S. Liu, P. Zhou, W. Hu, F. Xiu, *Adv. Funct. Mater.* **2015**, 25, 4076.
- [22] J. S. Ross, P. Klement, A. M. Jones, N. J. Ghimire, J. Yan, D. G. Mandrus, T. Taniguchi, K. Watanabe, K. Kitamura, W. Yao, D. H. Cobden, X. Xu, *Nat. Nanotechnol.* **2014**, 9, 268.
- [23] O. Salehzadeh, N. H. Tran, X. Liu, I. Shih, Z. Mi, *Nano Lett.* **2014**, 14, 4125.
- [24] Z. Yin, X. Zhang, Y. Cai, J. Chen, J. I. Wong, Y.-Y. Tay, J. Chai, J. Wu, Z. Zeng, B. Zheng, H. Y. Yang, H. Zhang, *Angew. Chem. Int. Ed.* **2014**, 53, 12560.
- [25] A. Pospischil, M. M. Furchi, T. Mueller, *Nat. Nanotechnol.* **2014**, 9, 257.
- [26] L. Niu, K. Li, H. Zhen, Y.-S. Chui, W. Zhang, F. Yan, Z. Zheng, *Small* **2014**, 10, 4651.

- [27] M.-L. Tsai, S.-H. Su, J.-K. Chang, D.-S. Tsai, C.-H. Chen, C.-I. Wu, L.-J. Li, L.-J. Chen, J.-H. He, *ACS Nano* **2014**, *8*, 8317.
- [28] Y.-H. Chang, W. Zhang, Y. Zhu, Y. Han, J. Pu, J.-K. Chang, W.-T. Hsu, J.-K. Huang, C.-L. Hsu, M.-H. Chiu, T. Takenobu, H. Li, C.-I. Wu, W.-H. Chang, A. T. S. Wee, L.-J. Li, *ACS Nano* **2014**, *8*, 8582.
- [29] M. Chhowalla, H. S. Shin, G. Eda, L.-J. Li, K. P. Loh, H. Zhang, *Nat. Chem.* **2013**, *5*, 263.
- [30] Z. Yin, H. Li, H. Li, L. Jiang, Y. Shi, Y. Sun, G. Lu, Q. Zhang, X. Chen, H. Zhang, *ACS Nano* **2012**, *6*, 74.
- [31] D. Ovchinnikov, A. Allain, Y.-S. Huang, D. Dumcenco, A. Kis, *ACS Nano* **2014**, *8*, 8174.
- [32] N. Perea-Lopez, A. L. Elias, A. Berkdemir, A. Castro-Beltran, H. R. Gutierrez, S. Feng, R. Lv, T. Hayashi, F. Lopez-Urias, S. Ghosh, B. Muchharla, S. Talapatra, H. Terrones, M. Terrones, *Adv. Funct. Mater.* **2013**, *23*, 5511.
- [33] J.-K. Huang, J. Pu, C.-L. Hsu, M.-H. Chiu, Z.-Yu. Juang, Y.-H. Chang, W.-H. Chang, Y. Iwasa, T. Takenobu, L.-J. Li, *ACS Nano* **2014**, *8*, 923.
- [34] W. Zhang, M.-H. Chiu, C.-H. Chen, W. Chen, L.-J. Li, A. T. S. Wee, *ACS Nano* **2014**, *8*, 8653.
- [35] S. Yang, S. Tongay, Y. Li, Q. Yue, J.-B. Xia, S.-S. Li, J. Li, S.-H. Wei, *Nanoscale* **2014**, *6*, 7226.
- [36] X. Wang, L. Huang, Y. Peng, N. Huo, K. Wu, C. Xia, Z. Wie, S. Tongay, J. Li, *Nano Res.* **2016**, *9*, 507.
- [37] S. Yang, C. Wang, H. Sahin, H. Chen, Y. Li, S.-S. Li, A. Suslu, F. M. Peeters, Q. Liu, J. Li, S. Tongay, *Nano Lett.* **2015**, *15*, 1660.
- [38] D. Wolverson, S. Crampin, A. S. Kazemi, A. Illi, S. J. Bending, *ACS Nano* **2014**, *8*, 11154.
- [39] P. Tonndorf, R. Schmidt, P. Bottger, X. Zhang, J. Borner, A. Liebig, M. Albrecht, C. Kloc, O. Gordan, D. R. T. Zahn, S. M. de Vasconcellos, R. Bratschitsch, *Opt. Exp.* **2013**, *21*, 4908.
- [40] A. Rai, A. Valsaraj, H. C. P. Movva, A. Roy, R. Ghosh, S. Sonde, S. Kang, J. Chang, T. Trivedi, R. Dey, S. Guchhait, S. Larentis, L. F. Register, E. Tutuc, S. K. Banerjee, *Nano Lett.* **2015**, *15*, 4329.
- [41] K. Chen, D. Kiriya, M. Hettick, M. Tosun, T.-J. Ha, S. R. Madhupathy, S. Desai, A. Sachid, A. Javey, *APL Mater.* **2014**, *2*, 092504.
- [42] B. Chakraborty, A. Bera, D. V. S. Muthu, S. Bhowmick, U. V. Waghmare, A. K. Sood, *Phys. Rev. B* **2012**, *85*, 161403.
- [43] J. D. Lin, C. Han, F. Wang, R. Wang, D. Xiang, S. Qin, X.-A. Zhang, L. Wang, H. Zhang, A. T. S. Wee, W. Chen, *ACS Nano* **2014**, *8*, 5323.
- [44] H.-Y. Park, M.-H. Lim, J. Jeon, G. Yoo, D.-H. Kang, S. K. Jang, M. H. Jen, Y. Lee, J. H. Cho, G. Y. Yeom, W.-S. Jung, J. Lee, S. Park, S. Lee, J.-H. Park, *ACS Nano* **2015**, *9*, 2368.
- [45] D.-H. Kang, J. Shim, S. K. Jang, J. Jeon, M. H. Jeon, G. Y. Yeom, W.-S. Jung, Y. H. Jang, S. Lee, J.-H. Park, *ACS Nano* **2015**, *9*, 1099.
- [46] H.-Y. Park, S. R. Dugasani, D.-H. Kang, J. Jeon, S. K. Jang, S. Lee, Y. Roh, S. H. Park, J.-H. Park, *ACS Nano* **2014**, *8*, 11603.
- [47] K. F. Mak, C. Lee, J. Hone, J. Shan, T. F. Heinz, *Phys. Rev. Lett.* **2010**, *105*, 136805.
- [48] D.-S. Tsai, K.-K. Liu, D.-H. Lien, M.-L. Tsai, C.-F. Kang, C.-A. Lin, L.-J. Li, J.-H. He, *ACS Nano* **2013**, *7*, 3905.
- [49] D. J. Groenedijk, M. Buscema, G. A. Steele, S. M. de Vasconcellos, R. Bratschitsch, H. S. J. van der Zant, A. Castellanos-Gomez, *Nano Lett.* **2014**, *14*, 5846.
- [50] V. Paril, A. Capone, S. Strauf, E.-H. Yang, *Sci. Rep.* **2013**, *3*, 2791.
- [51] W. Zhang, J.-K. Huang, C.-H. Chen, Y.-H. Chang, Y.-J. Cheng, L.-J. Li, *Adv. Mater.* **2013**, *25*, 3456.
- [52] M. M. Furchi, D. K. Polyushkin, A. Pospischil, T. Mueller, *Nano Lett.* **2014**, *14*, 6165.
- [53] N. R. Pradhan, J. Ludwig, Z. Lu, D. Rhodes, M. M. Bishop, K. Thirunavukkuarasu, S. A. McGill, D. Smirnov, L. Balicas, *ACS Appl. Mater. Interfaces* **2015**, *7*, 12080.
- [54] G. Su, V. G. Hadjiev, P. E. Loya, J. Zhang, S. Lei, S. Maharjan, P. Dong, P. M. Ajayan, J. Lou, H. Peng, *Nano Lett.* **2015**, *15*, 506.
- [55] D.-H. Kang, M.-S. Kim, J. Shim, J. Jeon, H.-Y. Park, W.-S. Jung, H.-Y. Yu, C.-H. Pang, S. Lee, J.-H. Park, *Adv. Funct. Mater.* **2015**, *25*, 4219.
- [56] Y. Zhang, V. W. Brar, C. Girit, A. Zettl, M. F. Crommie, *Nat. Phys.* **2009**, *5*, 722.
- [57] Y.-C. Wu, C.-H. Liu, S.-Y. Chen, F.-Y. Shih, P.-H. Ho, C.-W. Chen, C.-T. Liang, W.-H. Wang, *Sci. Rep.* **2015**, *5*, 11472.